MIL-C-81933(AS)
23 March 1973
Superseding AS-1068
25 January 1967

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 129 MOD 0

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

This document governs the manufacture, assembly and preparation for delivery of the Cartridge, Impulse, MARK 129 MOD 0 and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-P-20306	Powder, Ignition Cartridge, (M9), 60MM and 81MM Mortar
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-46610	Primers, Percussion, Styphnate and Chlorate Types, for Small Arms Ammunition

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 1377

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MIL-STD-414 Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS AND ASSOCIATED LISTS

Naval Air Systems Command (Code Identification No. 30003)

DL 2519638 Cartridge, Impulse, MARK 129 MOD 0 (Formerly Code Identification No. 10001)

DL 2519674 Test Fixture, Cartridge, Impulse, MARK 129 MOD 0

2519638 Cartridge, Impulse, MARK 129 MOD 0, Assembly (Formerly Code Identification No. 10001)

2519667 Cartridge, Impulse, MARK 129 MOD 0 Packaging Assembly (Formerly Code Identification No. 10001)

2519668 Container, Inner, Assembly (Formerly Code Identification No. 10001)

U. S. Army Munitions Command (Code Identification No. 19200)

C-7645332 Primer, Percussion (120M-257W)

PUBLICATIONS

Bureau of Naval Weapons (Code Identification No. 10001)

OP 5 Ammunition Ashore (This publication is not furnished to bidders and contractors by the procuring activity. Contractors should contact the Government Inspector for use of the publication.)

WR-43 Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178 Transportation, Explosives and Other Dangerous Articles

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General

3.1.1 The MARK 129 MOD 0 impulse cartridge covered by this specification shall be manufactured in accordance with the drawings listed on DL 2519638.

3.2 Materials. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referenced on the respective NAVAIR drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy cognizant field activity for cartridges prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of seventy-six MARK 129 MOD 0 impulse cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purposes of this specification, the primer and the main propellant charge are defined as primary components. See 4.3.3.

3.4 Primer. The primers shall be furnished from lots which were manufactured within the 12-month period immediately preceding the assembly of the primers into cartridges. The primers shall be manufactured in accordance with Drawing C-7645332, loaded with Olin Industries Primer Mixture 257W; and shall be from lots which were accepted in accordance with the applicable requirements of MIL-P-46610, Class 1. See 4.7.1.

3.5 Propellant charges.

3.5.1 Type and quantity. All propellant charges shall be as specified on Drawing 2519638. Charges shall be determined by weight rather than by volume measurements.

3.5.2 Smokeless powder, handling and loading. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

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3.5.3 Smokeless powder, condition. The smokeless powder utilized shall exhibit no evidence of decomposition. See 4.7.2.

3.6 Cartridge inspection.

3.6.1 Visual inspection. Cartridges shall be free of the following visible defects: dents, deep scratches, cracks, burrs, damaged "0" ring, incomplete or defective soldering, excessive solder, or any other imperfections which could prevent entry into the firing chamber or adversely affect performance or resistance to moisture. See 4.7.3.

3.6.2 Leakage. Cartridge leakage rate shall not exceed 10^{-5} cubic centimeters of air or gas per second when tested in accordance with 4.7.4.

3.7 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.7.5, shall meet the requirements listed below when subjected to the test prescribed in 4.7.5.

3.7.1 Chamber pressure. The lower (L) and upper (U) limits of maximum pressure in pounds per square inch shall be 3730 and 4060, respectively.

3.7.2 Misfire. There shall be no misfires. See 4.7.5.1.2.

3.7.3 Hangfire. There shall be no hangfires. See 4.7.5.1.3.

3.8 Change in propellant lot. Whenever a change occurs in propellant lot, test firings shall be conducted in accordance with 4.7.7. Certification of the new lot's conformance to the requirements of MIL-P-20306 shall be required.

3.9 Packaging, packing and marking. The packaging and packing of cartridges and the container markings shall be in accordance with Section 5. See 4.6.1.

3.10 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be directed to dimensions, finishes and coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105 or MIL-STD-414, whichever applies.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation and size shall be in accordance with MIL-STD-105 or MIL-STD-414, whichever applies.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 76 cartridges shall be prepared. Of these cartridges, 66 shall be expended in the tests listed below and 10 shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.3 through 3.6.2 of this specification. Preproduction sample acceptance shall consist of the following tests as prescribed in Table 3 of MIL-D-21625: 40-foot drop (6) (nonfiring test); 15 "g" shock (12); salt spray (6); temperature and humidity cycling (18); vibration (12); normal temperature (untreated) (12). The numbers in parentheses denote the allocation of cartridges for the tests. After exposure to the specified environmental treatment, these cartridges, except the 40-foot drop cartridges, shall be conditioned for a minimum of six hours at $+70^{\circ} \pm 5^{\circ}F$ and fired with 5 minutes after removal from conditioning. All preproduction cartridge firings shall be conducted in a test fixture in accordance with the requirements of 4.7.5. All cartridges shall fire and no cartridge shall produce a hangfire as defined in 4.7.5.1.3. Additionally, the lower (L) and upper (U) limits of the resultant maximum chamber pressures in pounds per square inch shall be L = 3510 and U = 4060, except that the pressures produced by the normal temperature (untreated) cartridges shall be within the limits specified in 3.7.1. The failure of any cartridge to meet the applicable requirements as stated above shall cause rejection of the preproduction sample.

4.3.3 Production lot. Unless otherwise specified in the contract, the minimum and maximum production lot size shall be 301 and 1300 cartridges, respectively. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.4 Gages. The contractor shall provide gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C), and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

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4.6 Quality conformance inspection. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL 2519638, and verification of the following requirements and tests:

Item	Nature of Requirement Verification	Applicable Paragraphs	Classification (WR-43)
Primary component	Single lot	3.4, 4.3.3	(M101)
Primer	Age, loading and acceptance	3.4, 4.7.1	(M102)
Propellant charges	Type and weight	3.5.1	(M103)
Smokeless powder	Condition	3.5.3, 4.7.2	(M104)
Cartridge	Visual inspection	3.6.1, 4.7.3	(M105)
Cartridge	Leakage	3.6.2, 4.7.4	(M106)
Cartridge	Ballistic test	3.7, 4.7.5	(M107)
Cartridge	Ballistics (change in propellant lot)	3.8, 4.7.7	(M108)
Cartridge	Packaging, packing and container markings	3.9, 4.6.1	(M109)

4.6.1 Packaging, packing and marking. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in Section 5.

4.7 Tests.

4.7.1 Primer verification. The contractor shall provide verification of primer conformity with the requirements of 3.4.

4.7.2 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP-5. Evidence of decomposition shall be cause for rejection.

4.7.3 Visual inspection. Examine all completely assembled cartridges and reject those having any of the defects noted in 3.6.1.

4.7.4 Leakage. Each cartridge in the preproduction sample or production lot ballistic sample shall be leak tested in a dry gas or dry air leak tester. Cartridges which exhibit a leak rate in excess of 10^{-5} cubic centimeters of air or gas per second at a pressure differential of 1.0 ± 0.1 atmosphere shall be considered defective. Cartridges of the preproduction sample which exhibit a leakage defect shall be rejected. If one or more cartridges of the production lot sample exhibit a leakage defect, then a 100 percent leak test of the lot being tested shall be required at no additional expense to the Government. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected.

4.7.5 Ballistic tests. The number of cartridges selected from each production lot for the ballistic tests shall be in accordance with the requirements of Table A-2 of MIL-STD-414, Inspection Level IV. The cartridges shall be conditioned for a

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minimum of six hours at +70° ± 5°F and fired in a test device fabricated in accordance with the drawings listed on DL 2519674. Each cartridge shall be fired within five minutes after removal from conditioning. Instrumentation shall be provided to measure the maximum chamber pressure and the time interval between the actuation of the firing mechanism and the start of pressure rise. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from the production lot for investigative purposes.

4.7.5.1 Criteria of acceptability.

4.7.5.1.1 Chamber pressure. Lot acceptability from the standpoint of the maximum chamber pressures produced by the sample shall be determined in accordance with the applicable provisions of MIL-STD-414, using an AQL of 1.0 and the pressure limits specified in 3.7.1.

4.7.5.1.2 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.7.5.1.3 Hangfire. One or more hangfires shall result in rejection of the lot represented. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and the start of pressure rise exceeds 0.05 second.

4.7.6 Retest. There shall be no retests.

4.7.6.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and that part of the test reconducted.

4.7.7 Change in propellant lot. Whenever a change in propellant lot occurs, the validity of the charge weight shall be verified by firing 36 cartridges which have been temperature conditioned as follows: 12 at -65° ± 5°F, 12 at +70° ± 5°F and 12 at +160° ± 5°F. Except as otherwise specified in this paragraph, the temperature conditioning and firing of these cartridges and firing results shall be in accordance with the requirements specified in 4.3.2 for the preproduction sample. The cartridge conditioning time at -65° and at +160° shall be for a period of not less than six nor more than twenty-four hours and the cartridges shall be fired within three minutes after removal from conditioning. The maximum chamber pressure produced by the cartridges conditioned at -65° and at +160° shall be within the limits specified in 4.3.2 for the cartridges subjected to environmental treatment. The maximum chamber pressure produced by the cartridges conditioned at +70° shall be within the limits specified in 3.7.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

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5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Twenty cartridges shall be packaged in an inner container in accordance with Drawing 2519668.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests, shall be packaged in accordance with Method III of MIL-P-116, and with the Code of Federal Regulations 49 CFR 171-178.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Eight inner containers as described in 5.1.1 shall be packed in accordance with Drawing 2519667.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests, and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3.2 Normal markings. Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on Drawings 2519668 and 2519667, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 Intended use. The MARK 129 MOD 0 impulse cartridge provides the power to operate the MARK 16 MOD 0 minesweeping cable cutter.

6.2 Ordering information. Procurement documents should specify the following:

a. Title, number and date of this specification;

b. whether preproduction inspection is required and, if so, specify the test activity;

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c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification, or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order.

6.3.1 Test report. The contract or order should specify that results of ballistic tests are to be forwarded to the Naval Air Systems Command (AIR-53222A) and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. For the purposes of this specification, the following definitions apply.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level, in general, will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 Explosives safety precautions. Safety precautions for explosive-loaded items are contained in DOD 4145.26M, "DOD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material".

Project No. 1377-N517

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-81933(AS) CARTRIDGE, IMPULSE, MARK 129 MOD 0

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT. 64 WHICH MAY BE USED.

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Washington, D. C. 20361

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